CV

AI CAD-to-Cost Analysis

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AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

47

Fuel_tank

1528 × 658 × 594 mm · 10595.9 cm³ · Al 28.609 kg / Steel 83.178 kg

Manufacturability: 47/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	**1528.3 × 657.6 × 593.7 mm**	Surface Area	**47571.3 cm²**
Volume	**10595.90 cm³**	Weight (Al)	**28.609 kg**
Weight (Steel)	**83.178 kg**	Weight (Plastic)	**11.126 kg**

§2 — Detected Features

Feature Type	Count	Significance	Description
Free-form shell body	694	High	694 B-spline surfaces forming the organic saddle/underbody tank shape typical of automotive fuel tanks
Boss/fitting features	12	High	Cylindrical bosses (r=12,15,26,40mm) for filler neck, vent valve, and pump-module flange
Mounting/vent holes	8	Medium	Holes at r=8,20,30mm for straps, grommets and vent lines
Threaded ports	1	Medium	Threaded features detected — likely pump module lock-ring or fitting threads, typically post-moulded inserts on tanks
Pinch-weld thickened zones	1	High	Max wall 18.34mm vs mean 5.04mm indicates parison pinch-off/weld lines at parting plane

§3 — Material Analysis

AI-suggested

Primary Material: **HDPE (6-layer barrier grade) (80% confidence)**

Part is a large automotive fuel tank (bounding box 1.5m, hollow shell, fill ratio 0.0178, wall 4-18mm). HDPE with EVOH barrier is the industry-standard resin for extrusion blow moulded fuel/AdBlue tanks requiring hydrocarbon permeation barrier; PP and PET are not used for fuel containment. Resin selected from part function/geometry, not from title-block metadata.

Alternatives: **PP Homopolymer (MFI 12) (8%) · PA6 (barrier alternative) (5%)**

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Extrusion Blow Moulding (6-layer barrier)	blow_moulding	92%	0.0250	USER-FORCED process. Large hollow single-piece tank with 5mm nominal wall, thick pinch-weld zones (up to 18.34mm), and 753 undercuts consistent with a shuttle-type multilayer EBM machine using accumulator-head parison programming and blow-pin/side-action tooling for the filler neck and pump flange.

§5 — Manufacturability Risks (Score: 47/100)

Severity	Feature / Area	Description	Recommended Action
High	**753 undercut faces**	Numerous undercuts (fittings, baffle bosses, filler neck) require blow-pin side actions, split-cavity inserts and precise pinch-off alignment, increasing mould complexity and risk of witness-line leak paths.	Consolidate undercut features onto the parting plane where possible; use retractable core pins for filler neck/vent bosses rather than full side-action inserts.
Medium	**Wall thickness variation (Ã = 2.82mm, 4.15–18.34mm)**	Wide wall variation between thin body sections and thick pinch welds causes uneven cooling, sink, and warpage risk, and extends cycle time governed by the thickest section.	Programme parison wall-thickness profile (die-head parison control) to equalise draw-down and minimise pinch-weld bulk.
Medium	**Multilayer barrier uniformity**	6-layer coextrusion (HDPE/tie/EVOH/tie/HDPE) must maintain continuous EVOH barrier layer through pinch welds and around bosses; local thinning risks HC permeation failure.	Verify layer distribution via cross-section sampling near pinch welds and boss features during PPAP.
Low	**Free-form organic surfaces (694 B-spline faces)**	Complex packaging-driven contour increases mould machining time and CMM inspection complexity.	Use 5-axis mould cavity machining and laser-scanning inspection fixtures.

§6 — DFM Issues (blow_moulding)

Severity	Area	Description	Impact	Fix
High	**Pinch weld / parting line**	Max wall 18.34mm vs 5.04mm mean indicates thick pinch-off zones where parison material doubles up; these are common failure initiation sites under fuel-vapour pressure cycling.	Risk of weld-line leaks in service, scrap/rework at trim stage, and warranty exposure.	Optimise parison programming and pinch-off land geometry; add rib reinforcement at pinch-weld line.
High	**Undercut bosses (753 faces)**	Filler neck, vent, and pump-module bosses create undercuts relative to the blow axis, requiring side-action blow pins or split inserts.	Increases mould cost (~15-20%) and cycle time; risk of flash/witness marks at fitting interfaces affecting sealing.	Design bosses to be moulded in-line with draw direction or add them as post-moulded welded fittings (spin-weld/vibration-weld domes).
Medium	**Barrier layer continuity**	6-layer HDPE/EVOH coextrusion must maintain a continuous EVOH barrier through thickness transitions and pinch welds.	Non-uniform barrier layer can fail hydrocarbon-permeation regulatory limits (evap emissions).	Increase EVOH layer ratio locally at pinch weld tooling or add a barrier reinforcement patch.
Medium	**Thin wall near bosses (min 4.15mm)**	Local wall thinning at boss root during blow stretch increases blow-out and porosity risk.	Higher scrap rate, especially at higher line speed for the 200k/yr volume target.	Increase local parison thickness profile at boss locations or add pre-blow assist.

Severity	Area	Description	Impact	Fix
Low	**Large single-parison size**	1.5m long parison for a single-shot tank is prone to sag/drawdown non-uniformity top-to-bottom.	Wall thickness scatter reduces structural consistency and adds inspection burden.	Use multi-die-head parison thickness profiling and a vertical shuttle mould to minimise sag time.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£13.00	£17.50	£23.00

Net Weight	**11.126 kg**	Material	**mat-hdpe**
Recommended Process	**blow_moulding**	Cycle Time	**0.0250 hr/part**
Setup Time	**2.000 hr**	Operations	**0 ops**
Operations Detail			

Process-Specific Parameters

Casting Subtype	**sand**
Die/Mould Cost	**£50,000**
Die Life	**1,000 shots**
Cavities	**1**
Yield	**85.0%**
Flash Weight	**0.000 kg**
Yield	**80.0%**
Die Cost	**£180,000**
Strokes	**8**
Cavities	**1**
Wall Thickness	**5 mm**
Mould Cost	**£200,000**
Mould Life	**500,000 shots**
Projected Area	**10046.0 cm²**

§8 — AI Analysis Explanation

The geometry (1.53m x 0.66m x 0.59m hollow shell, fill ratio 0.0178, wall 4.15-18.34mm mean 5.04mm, 753 undercuts, 694 free-form faces) is characteristic of an automotive extrusion-blow-moulded fuel tank. The user-forced process (blow_moulding) and part shape/function point to HDPE 6-layer barrier resin rather than PP or PET, since fuel tanks require EVOH hydrocarbon barrier and structural stiffness that PET/PP cannot provide at this size. Deterministic OCCT values (wall thickness, volume, weights, manufacturability score, tooling cost baselines) were used verbatim; UNDECIDED blow-moulding cost fields (material, subtype, flash, cavities, mould cost/life, blow/open-close times, barrier flag) were estimated from part function and typical large-tank EBM practice, each flagged with a lower confidence score.

§9 — Analysis Limitations & Assumptions

1. No blow-moulding-specific machineId exists in the valid machine list; process selection and cycle time are geometry-derived estimates.
2. Flash weight, mould cost/life, and blow/open-close timing are typical-practice estimates, not derived from supplier quotes.
3. Cavities assumed at 1 due to large part size; multi-cavity shuttle tooling not modelled.

Stage-1 pre-selection: auto (0%) ·